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Hierarchical Algorithms
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Exercise sheet 08, Solution hints
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Homework:

Exercise 1: Let $d, n \in \mathbb{N}$ and let $\{x_0, \dots, x_n\} \subset \mathbb{R}^d$ be a point set corresponding to the index set $I = \{0, \dots, n\}$. A cluster tree T corresponding to the index set I can be constructed geometry-based as follows:

1. Set $r := I$ as the root of T.
2. Compute the smallest cuboid $Q_r = \bigtimes_{i=1}^{d} [a_i, b_i] \subset \mathbb{R}^d$ that includes the points that are assigned to the indices in r.
3. Determine a direction $k \in \{1, \dots, d\}$ with $b_k - a_k = \max\{b_i - a_i : i \in \{1, \dots, d\}\}$.
4. Compute $m_k = \frac{a_k + b_k}{2}$, split Q_r into the two cuboids

$$Q_{r_1} = \bigtimes_{i=1}^{k-1} [a_i, b_i] \times [a_k, m_k] \times \bigtimes_{i=k+1}^{d} [a_i, b_i],$$
$$Q_{r_2} = \bigtimes_{i=1}^{k-1} [a_i, b_i] \times [m_k, b_k] \times \bigtimes_{i=k+1}^{d} [a_i, b_i],$$

   and divide r corresponding to the index sets r_1, r_2 where

$$r_1 := \{j \in r : x_j \in Q_{r_1}\} \text{ und } r_2 := r \setminus r_1.$$

5. For $i \in \{1, 2\}$, execute the steps 3., 4. and 5. for the cuboid Q_{r_i}, the index set r_i and the corresponding points if $\#r_i > 1$. This creates the next level in the cluster tree T.

Proof the following statements for the cluster tree T created by this algorithm:

a) The described algorithm terminates. The created cluster tree satisfies $\#S(t) \leq 2$ for all $t \in T$ where (the typically undesired case) $\#S(t) = 1$ may occur.

b) It holds

$$\operatorname{depth}(T) \leq d \cdot \left(\left\lfloor \log_2 \left(\frac{\delta_{\max}}{\delta_{\min}} \right) \right\rfloor + 1 \right)$$

   for the depth of the cluster tree with $\delta_{\min} := \min\{\|x_i - x_j\|_\infty \mid i, j \in I, i \neq j\}$ and $\delta_{\max} := \max\{\|x_i - x_j\|_\infty \mid i, j \in I\}$.

In the algorithm above, step 5. is replaced by the following step 5.'

5.' For $i \in \{1, 2\}$ perform step 2. to compute the smallest cuboid Q'_{r_i} that includes the points assigned to the indices in r_i and perform the steps 3., 4. and 5. for the cuboid Q'_{r_i}, the index set r_i, and the corresponding points if $\#r_i > 1$. This creates the next level in the cluster tree T'.

Proof the following statements for the cluster tree T' created by the modified algorithm:

c) For all $t' \in T'$, $\#S(t') \in \{0, 2\}$ holds.

d) We hope that $\operatorname{depth}(T') \leq \operatorname{depth}(T)$ holds for the depth of the cluster tree T'. However, there exist counterexamples. Think of an example (i.e. a particular point set $\{x_0, \ldots, x_n\} \subset \mathbb{R}^d$) such that $\operatorname{depth}(T') < \operatorname{depth}(T)$ or $\operatorname{depth}(T') > \operatorname{depth}(T)$, respectively, holds.

Solution hints:

a) The index set I is finite and in step 4., we determine a partioning of the index set I into two index sets. Since the recursion in step 5. is only called if at least one of the index sets contains more than one index, the algorithm terminates.

In step 4., the index set r is always divided into two index sets r_1, r_2. Hence, $\#S(t) \leq 2$ holds for all $t \in T$ since each cluster can then have at most two successors. The construction of Q_{r_1}, Q_{r_2} does not exclude that one of the index sets r_1, r_2 is empty. If $r_1 = \emptyset$ or $r_2 = \emptyset$ then we obtain $\#S(r) = 1$.

b) Let $Q_{\text{level}(d)}$ be a cuboid created from d divisions of Q_r, i.e. $Q_{\text{level}(d)}$ corresponds to a cluster on level d of the cluster tree T. Thus, the following applies

$$\operatorname{diam}_\infty(Q_{\text{level}(d)}) \leq \frac{1}{2} \operatorname{diam}_\infty(Q_r) = \frac{1}{2} \max\{b_i - a_i : i \in \{1, \ldots, d\}\} = \frac{1}{2} \delta_{\max},$$

since $Q_r = \bigtimes_{i=1}^d [a_i, b_i] \subset \mathbb{R}^d$ is the smallest cuboid that includes the points corresponding to the indices in r.

Let Q_s be a cuboid created from Q_r that satisfies

$$\operatorname{diam}_\infty(Q_s) < \delta_{\min}.$$

Then, $\#Q_s \leq 1$ holds. From

$$\begin{aligned} \operatorname{diam}_\infty(Q_{\text{level}(d \cdot k)}) &\leq \left(\frac{1}{2}\right)^k \delta_{\max} \overset{!}{<} \delta_{\min} \\ &\Rightarrow \frac{1}{2^k} < \frac{\delta_{\min}}{\delta_{\max}} \\ &\Rightarrow k > \log_2\left(\frac{\delta_{\max}}{\delta_{\min}}\right) \end{aligned}$$

we obtain that $\operatorname{depth}(T) \leq d \cdot k$ holds with

$$k := \left\lfloor \log_2\left(\frac{\delta_{\max}}{\delta_{\min}}\right) \right\rfloor + 1.$$

c) In a) we already showed that $\#S(t) \leq 2$ for all $t \in T$. Analogously, we obtain $\#S(t') \leq 2$ for all $t' \in T'$. The construction of Q'_{r_1}, Q'_{r_2} together with step 3. excludes that one of the index sets r_1, r_2 can be empty. We obtain $\#S(r) \neq 1$.

d) For the point set $\{0, 1, 5\} \subset \mathbb{R}$ holds $\operatorname{depth}(T') = 2 < 3 = \operatorname{depth}(T)$. For this case, a vertex $t \in T$ with $S(t) = 1$ exists.
For the point set $\{0, \frac{1}{2} - \frac{1}{32}, \frac{1}{2} + \frac{1}{32}, \frac{3}{4} + \frac{1}{8}, 2\} \subset \mathbb{R}$, we obtain the following cluster trees with $\operatorname{depth}(T') = 4 > 3 = \operatorname{depth}(T)$:

T

	$\{0, \frac{1}{2} - \frac{1}{32}, \frac{1}{2} + \frac{1}{32}, \frac{3}{4} + \frac{1}{8}, 2\}$			Division at 1
	$\{0, \frac{1}{2} - \frac{1}{32}, \frac{1}{2} + \frac{1}{32}, \frac{3}{4} + \frac{1}{8}\}$		$\{2\}$	Division at $\frac{1}{2}$
$\{0, \frac{1}{2} - \frac{1}{32}\}$		$\{\frac{1}{2} + \frac{1}{32}, \frac{3}{4} + \frac{1}{8}\}$		Division at $\frac{1}{4}$ or $\frac{3}{4}$
$\{0\}$	$\{\frac{1}{2} - \frac{1}{32}\}$	$\{\frac{1}{2} + \frac{1}{32}\}$	$\{\frac{3}{4} + \frac{1}{8}\}$	

T'

	$\{0, \frac{1}{2} - \frac{1}{32}, \frac{1}{2} + \frac{1}{32}, \frac{3}{4} + \frac{1}{8}, 2\}$			Division at 1
	$\{0, \frac{1}{2} - \frac{1}{32}, \frac{1}{2} + \frac{1}{32}, \frac{3}{4} + \frac{1}{8}\}$		$\{2\}$	Division at $\frac{1}{2}(\frac{3}{4} + \frac{1}{8} + 0) = \frac{7}{16} = \frac{14}{32}$
$\{0\}$		$\{\frac{1}{2} - \frac{1}{32}, \frac{1}{2} + \frac{1}{32}, \frac{3}{4} + \frac{1}{8}\}$		Division at $\frac{1}{2}(\frac{3}{4} + \frac{1}{8} + \frac{1}{2} - \frac{1}{32}) = \frac{43}{64}$
	$\{\frac{1}{2} - \frac{1}{32}, \frac{1}{2} + \frac{1}{32}\}$		$\{\frac{3}{4} + \frac{1}{8}\}$	Division at $\frac{1}{2}$
	$\{\frac{1}{2} - \frac{1}{32}\}$	$\{\frac{1}{2} + \frac{1}{32}\}$		

Exercise 2: Let $d, n \in \mathbb{N}$ and let $\{x_0, \ldots, x_n\} \subset \mathbb{R}^d$ be a point set corresponding to the index set $I = \{0, \ldots, n\}$. A cluster tree T corresponding to the index set I can be constructed cardinality-based as follows:

1. Set $r := I$ as the root of T.
2. Compute the smallest cuboid $Q_r = \bigtimes_{i=1}^{d} [a_i, b_i] \subset \mathbb{R}^d$ that includes the points corresponding to the indices in r.
3. Determine a direction $k \in \{1, \ldots, d\}$ with $b_k - a_k = \max\{b_i - a_i : i \in \{1, \ldots, d\}\}$.
4. Sort the indices in $r = \{i_0, \ldots, i_n\}$ such that $x_{i_j,k} \leq x_{i_\ell,k}$ holds for all $j, \ell \in r$ with $j \leq \ell$. The coordinate $x_{i,k}$ with $i \in r$ and $k \in \{1, \ldots, d\}$ denotes the kth coordinate of the point x_i. Divide the index set r into the index sets r_1, r_2 where
$$r_1 := \{i_0, \ldots, i_{\lceil \#r/2 \rceil}\} \text{ und } r_2 := r \setminus r_1.$$
5. For $i \in \{1, 2\}$, perform the steps 2., 3., 4. and 5. for the index set r_i and the corresponding points if $\#r_i > 1$. This creates the next level of the cluster tree T.

Proof that this algorithm leads to a cluster tree T with
$$\operatorname{depth}(T) \leq \lceil \log_2(n+1) \rceil.$$

Solution hints:

For all $k \in \mathbb{N}$ with $k \leq \operatorname{depth}(T)$ and for all $t \in T$ with $\operatorname{level}(t) = k$,

$$\#t \leq \left\lceil \left(\frac{1}{2}\right)^k \cdot \#I \right\rceil$$

holds. If a vertex $t \in T$ with $\operatorname{level}(t) = \lceil \log_2(\#I) \rceil$ exists,

$$\#t \leq \left\lceil \left(\frac{1}{2}\right)^{\lceil \log_2(\#I) \rceil} \cdot \#I \right\rceil \leq \left\lceil \left(\frac{1}{2}\right)^{\log_2(\#I)} \cdot \#I \right\rceil = \lceil (2^{\log_2(\#I)})^{-1} \cdot \#I \rceil = \lceil (\#I)^{-1} \cdot \#I \rceil = 1$$

holds for this vertex. Hence, step 5. is not satisfied for the index set t and we obtain

$$\operatorname{depth}(T) \leq \lceil \log_2(\#I) \rceil = \lceil \log_2(n+1) \rceil.$$

Exercise 3: Let I be an index set, T be a cluster tree corrsponding to the index set I and $P \subset T$ be a partitioning of the index set I. We define $T(P)$ as the subtree of T that is constructed by "truncating" all branches below the cluster from P. $T(P)$ hence only includes the predecessors of the clusters in P, i.e. $\{t \in T : \exists p \in P \setminus \{r\} : t \in P^*(p)\}$ is the vertex set of $T(P)$.
Justify why the following statements hold for partitionings $P_1, P_2 \subset T$ (i.e. no formal proofs are expected)

a) $\min\{P_1, P_2\} := L(T(P_1) \cap T(P_2))$ is a partitioning of I with the properties

- $\min\{P_1, P_2\} \subset P_1 \cup P_2$.
- For all $b' \in P_1 \cup P_2, b \in \min\{P_1, P_2\}$ with $b' \cap b \neq \emptyset, b' \subset b$ holds .
- $\#\min\{P_1, P_2\} \leq \min\{\#P_1, \#P_2\}$.

b) $\max\{P_1, P_2\} := L(T(P_1) \cup T(P_2))$ is a partitioning of I with the properties

- $\max\{P_1, P_2\} \subset P_1 \cup P_2$.
- For all $b' \in P_1 \cup P_2, b \in \max\{P_1, P_2\}$ with $b' \cap b \neq \emptyset, b' \supset b$ holds .
- $\#\max\{P_1, P_2\} \geq \max\{\#P_1, \#P_2\}$.

Hint: The trivial partitionings are $P = \{r\}$ (root) and $P = L(T)$ (set of all leaves).
Solution hints:

The set of all leaves of a cluster tree is always a partitioning of the corresponding index set. $T(P_1)$ and $T(P_2)$ are cluster trees corresponding to the index set I. Hence, $T(P_1) \cap T(P_2)$ and $T(P_1) \cup T(P_2)$ are also cluster trees corresponding to the index set I. Hence, $\min\{P_1, P_2\}$ and $\max\{P_1, P_2\}$ are partitionings of I. The properties of the partitionings $\min\{P_1, P_2\}$ and $\max\{P_1, P_2\}$ arise immediately from the respective definition and mean that the partitioning $\min\{P_1, P_2\}$ is coarser and the partitioning $\max\{P_1, P_2\}$ is finer than P_1 and P_2.

Exercises (in class):

Exercise 4: In *cluster.h*, the structures `cluster` and `clustertree` are defined which are used to model a cluster and a cluster tree.

```
typedef struct _cluster cluster;
typedef cluster *pcluster;
typedef const cluster *pccluster;
struct _cluster {
  int start;
  int size;
  double *bmin;
  double *bmax;
  int d;
  int sons;
  pcluster *son;
};
```

The parameters `start` and `size` determine the smallest index and the number of indices of a cluster. The parameter `d` denotes the dimension of the corresponding points. The left and right boundary points of the intervals defining the cuboid that includes all assigned points are stored in `bmin` and `bmax`. The parameter `sons` denotes the number of successors of a cluster. The `sons` successors are stored in `son`.

```
typedef struct _clustertree clustertree;
typedef clustertree *pclustertree;
typedef const clustertree *pcclustertree;
struct _clustertree {
  int nidx;
  int *permute
  pcluster root;
};
```

`nidx` denotes the cardinality of the index set that corresponds to the cluster tree. The variable `permute` stores the renumbering of the points that arises in the construction of a cluster tree. The variable `root` stores the cluster that corresponds to the root of a cluster tree.

Familiarize yourself with the implementation of the geometry-based construction of a cluster tree corresponding to the index set $I = \{0, \ldots, n\}$ with point set $\{x_0, \ldots, x_n\} \subset \mathbb{R}^d$. In particular, take a look at the following methods in *cluster.c*

```
pcluster split_geometric_cluster(int start, int size, int d, const double
                                 *nodes_x, int nmin, int *permute);
pclustertree create_geometric_clustertree(int n, int d, const double *nodes_x,
                                          int nmin).
```

Test the cluster tree construction with *example_ 08a.c* for a relatively small number of vertices such that you understand the divisions and resulting permutations.
Solution hints:

Exercise 5: Familiarize yourself with the method

```
psupermatrix build_supermatrix_from_cluster(pccluster rc, pccluster cc,
                                            double eta);
```

implemented in *cluster.c* that constructs a $\mathcal{H}$-matrix-structure of two clusters `rc`, `cc` and the admissibility condition

$$\min\{\mathrm{diam}(rc),\ \mathrm{diam}(cc)\} \leq \eta\, \mathrm{dist}(rc, cc).$$

Perform tests for different parameter sets by using and adapting *example_ 08b.c* and *plot_ data.m* to visualize the results in MATLAB. Change for example the node set from a tensor grid to a randomly distributed point set (requires only a change of the uncommented parts) and vary the test functions (log, Gaussian, or any function of your choice). What changes regarding accuracy, computational times and memory requirements do you observe for increasing the spatial dimension?

Solution hints: